

CS 150: Artificial Agents and Autonomy Final Paper

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1 Introduction

This experiment is an attempt to find **generalizable** circuitry within the Gemma series of models for fact-finding using techniques from mechanistic interpretability i.e. activation patching using clean/corrupted prompt pairs. The specific models used are **Gemma-2b** and **Gemma-12b-it**, which have 12/48 layers (transformer blocks) and 8/16 attention heads respectively. All experimentation was done across three experimental models A, B, and C and the prompts themselves were triaged using the TotalSwing metric, all of which will be defined below.

When a language model states that the capital of Nepal is Kathmandu, something inside its forward pass located that fact, routed it to the right position, and made it readable at the output. Identifying precisely where and how that happens, which layers, which token positions, which attention heads, remains an open problem in mechanistic interpretability. The difficulty is not conceptual but surgical: transformer residual streams are high-dimensional, every layer writes to the same shared space, and correlation-based probing methods cannot establish causality. Activation patching addresses this by intervening directly, replacing one run's internal state with another's and measuring the consequence, but prior work has applied it narrowly, often to a single hook point or a single model, leaving the question of whether identified circuits are architectural universals or model-specific artifacts unresolved.

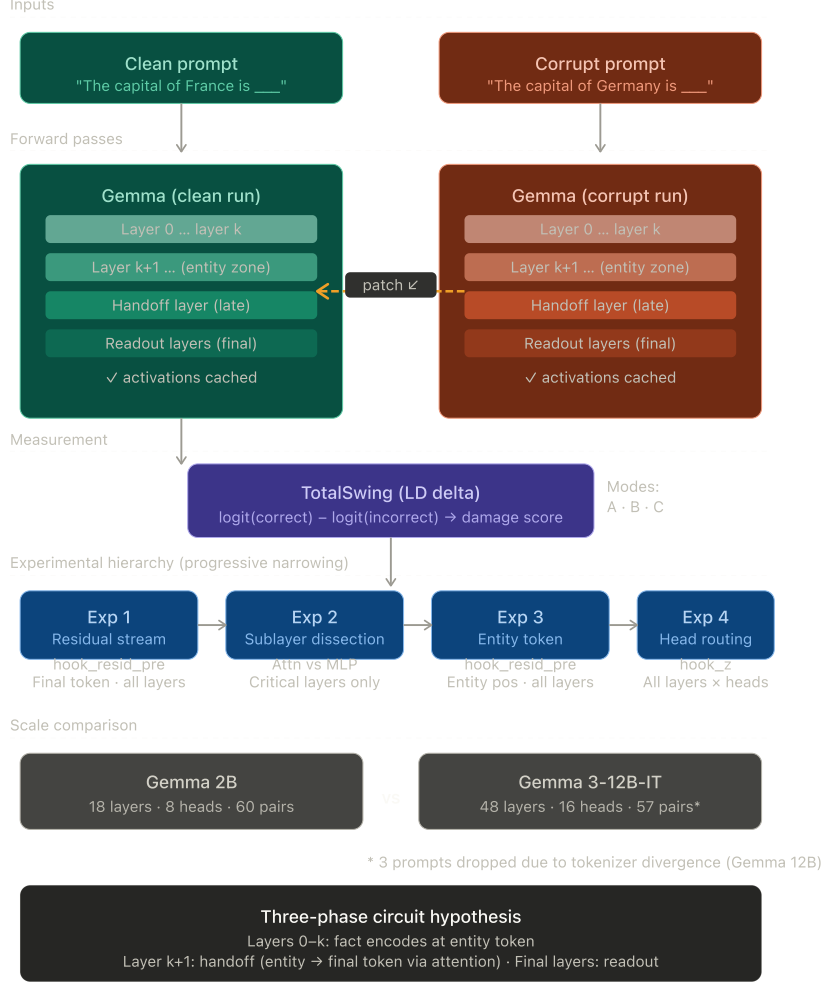


Figure 1: Overview of the BIZZAROWORLD activation patching pipeline. A clean and corrupt prompt pair are forward-passed through Gemma independently, with residual stream activations cached at every layer. A corrupt-run activation is surgically injected into the clean run at a target hook point and token position; the resulting drop in logit difference ($\text{TotalSwing} = \text{LD}_{\text{clean}} - \text{LD}_{\text{corrupt}}$) quantifies causal load at that site. Four experiments progressively narrow the circuit: (1) residual stream at the final token across all layers, (2) sublayer dissection (attention vs. MLP) at critical layers, (3) entity token position across all layers, and (4) individual attention heads via `hook_z`. All experiments run under three golden pair selection modes (A, B, C) and replicated on Gemma 3-12B-IT for cross-scale validation.

I present BizzaroWorld, a systematic activation patching study designed to causally identify the factual recall circuit in the Gemma model family. The approach constructs 60 matched prompt pairs across 20 knowledge categories, each pair differing in exactly one factual entity, and evaluates patching damage using a bidirectional logit difference metric (TotalSwing) that is robust to tokenization ambiguity and vocabulary distribution artifacts. Four experiments progressively narrow the circuit: residual stream patching at the final token position establishes which layers are load-bearing for readout; sublayer dissection separates attention and MLP contributions at those layers; entity token patching identifies where the fact encodes upstream; and head-level patching via $hook_z$ localizes the routing mechanism to individual attention heads. All experiments run under three independent golden pair selection strategies to guard against selection bias, and the full suite is replicated on Gemma12B-it to test whether the identified circuit scales with model depth.

Cross-scale results on Gemma-12b-it preserve the same qualitative layer-wise topology seen in Gemma-2b, but with a deeper timeline: early entity-patch damage is larger in magnitude (pooled layer-0 mean ΔLD : -18.55 vs -10.93) and release occurs substantially later (mean first layer with $\Delta LD \geq -1$: ~ 30.1 in 12b-it vs ~ 12.6 in 2b; zero-crossing typically in the mid-30s vs low-teens). This supports a scale-stable circuit topology with depth-stretched execution rather than a new mechanism. The trend is consistent across all three selection modes, with mode-to-mode variation in exact release layer but no mode showing a qualitatively different progression.

The approach constructs 60 matched prompt pairs across 20 knowledge categories, each pair differing in exactly one factual entity, producing a clean and a corrupt forward pass, and scores each pair using TotalSwing, a bidirectional logit difference metric that measures how strongly the model distinguishes the correct from the incorrect answer. Pairs are ranked by TotalSwing and drawn into three evaluation modes: Mode A selects the top 15 pairs globally, Mode B selects the strongest pair per category, and Mode C uses all 60 pairs. Running all experiments across all three modes allowed me to distinguish findings that are robust from those sensitive to pair selection. Four experiments then progressively narrowed the circuit: residual stream patching at the final token established which layers are load-bearing for readout; sublayer dissection isolated attention versus MLP contributions; entity token patching identified where the fact encodes upstream; and head-level patching via $hook_z$ localized the routing mechanism to individual attention heads. The full suite was replicated on Gemma-12B-it to test whether the identified circuit generalizes across model scale.

I find that factual recall in Gemma-2b is mediated by a three-phase circuit: information is causally encoded at the entity token in early-to-mid layers, a sharp handoff occurs in the low teens, and readout load concentrates near the final layers at the last-token position. Across modes A/B/C, the share of pairs releasing exactly at layer 15 is 80.0%, 65.0%, and 63.3% (66.3% pooled), indicating

a stable but not perfectly single-layer transition. This pattern is robust across pair-selection modes and remains strongly associated with confidence–damage relationships in our Exp1 analysis. Head-level analysis shows distributed routing, but with a clear bias toward head 2 (worst head in 47.4% of pooled pairs), suggesting a mixed mechanism: partially concentrated yet not exclusively dependent on one head.

Head-level analysis reveals distributed routing with no single dominant head, suggesting the handoff is a collective attention phenomenon. Preliminary cross-scale results on Gemma-12b-it show a proportionally shifted but structurally consistent circuit across 48 layers; three prompts were excluded due to tokenizer divergence, a cross-scale evaluation constraint we characterize in the methods.

2 Background

Activation patching was used as a causal tool, and the question of circuit stability across model variants together form the direct intellectual lineage of this work. I organize this section to move from foundational framing, through the methodological tools I adopt, to the most closely related prior empirical results, concluding with the gap the work addresses.

The mechanistic interpretability program, articulated by Olah et al. [5] and formalized mathematically by Elhage et al. [3], proposes that transformer language models implement identifiable, human-understandable algorithms in their weights. Rather than treating model behavior as an emergent property of the whole, this approach reverse-engineers the forward pass into *circuits*: sparse computational subgraphs of attention heads and MLP layers causally responsible for a given capability. The program is motivated by safety: a model whose internal algorithms can be verified is a model whose failure modes can be predicted. The core challenge is that individual neurons and attention heads are polysemantic: they encode multiple unrelated concepts simultaneously through superposition [2], making circuit boundaries diffuse and intervention-based methods necessary for causal, rather than merely correlational, claims.

Activation patching, also called causal tracing, provides the surgical instrument for this kind of analysis. The technique was applied at scale in Wang et al. [7], who reverse-engineered the indirect object identification (IOI) circuit in GPT-2 small. Given prompts of the form “When Mary and John went to the store, John gave a drink to,” the model must identify the non-repeated name. Wang et al. discovered a 26-head circuit organized into seven functional classes: Name Mover Heads, S-Inhibition Heads, Duplicate Token Heads, and others. The team validated the circuit against three quantitative criteria i.e. faithfulness, completeness, and minimality. The IOI work established the template for circuit analysis that subsequent studies have followed: construct matched prompt pairs, sweep causal interventions across layers and positions, and characterize the resulting heatmap of logit-difference damage. This project’s experimental design follows this template directly, adapting it from syntactic tasks to factual

recall across 20 knowledge categories.

Path patching, formalized by Goldowsky-Dill et al. [4], sharpens activation patching from node-level to edge-level precision. Standard activation patching measures the total causal effect of a node by replacing its output and observing all downstream consequences; path patching, instead, aggregates effects across multiple paths and can obscure the specific routing of information. This intercepts the activation of a sender node u only along a specific directed edge to a receiver node v , leaving all other outgoing paths from u intact.

Formally, if the transformer’s computational graph is a directed acyclic graph $G = (V, E)$, node (activation) patching tests the total effect of $u \in v$, while path patching tests the specific causal contribution of an edge $(u, v) \in E$. This is strictly more expressive and avoids a key confound: a head may causally contribute positively along one path and negatively along another, and node patching conflates both effects. BizzaroWorld’s four-experiment hierarchy is designed in the same spirit of progressive localization, moving from residual stream at the final token, to sublayer dissection, to entity token position, to individual attention heads via $hook_z$, each step narrowing the causal claim from the previous, and it could benefit from this kind of workflow.

The most directly relevant prior work is Prakash et al. [6], who investigate whether fine-tuning rewrites or merely amplifies the internal circuits of pre-trained models. Using entity tracking as the test task and Llama-7B as the base model, they identify a four-group attention head circuit using path patching and Desiderata-based Component Masking (DCM) [1], a gradient-based mask optimization technique that automates the discovery of which components encode specific semantic subtasks. They then introduce Cross-Model Activation Patching (CMAP), which patches activations not between inputs within one model but between the same architectural locations of different models on the same input. Their central finding is that fine-tuned variants (Vicuna, Goat, FLoat) share the same circuit as the base Llama-7B with faithfulness scores above 0.88, and that performance improvements are attributable to enhanced positional encoding in the Value Fetcher heads rather than new circuitry. This establishes an important prior: circuits are architecturally stable across post-training interventions within the same model family.

BizzaroWorld extends this line of inquiry in two directions that prior work does not address. First, whereas existing circuit analyses focus on syntactic or in-context reasoning tasks (IOI, entity tracking, induction), I study **factual recall**, a qualitatively different capability requiring retrieval of world knowledge encoded during pretraining rather than algorithmic processing of context structure. Second, no prior work applies a systematic, multi-experiment patching hierarchy to the Gemma family, nor examines whether the identified factual recall circuit is preserved as model depth scales from 18 layers (Gemma-2b) to 48 layers (Gemma-12b-it). The cross-scale comparison is distinct from CMAP, which patches across fine-tuned variants of the same architecture; we instead ask whether the *same circuit phases* appear at proportionally shifted layer indices

across a genuine depth increase, with a tokenizer divergence that itself surfaces as a methodological constraint on cross-scale comparability.

3 Results

I present the results in four stages, each progressively narrowing the causal circuit for factual recall. Experiments run across three golden pair selection modes (A, B, C) to guard against selection artifacts, and the full suite is replicated on Gemma-12b-it to assess cross-scale generalization.

3.1 Gemma 2B: Exp 1, 2, 3 (residual stream, sublayer, entity token)

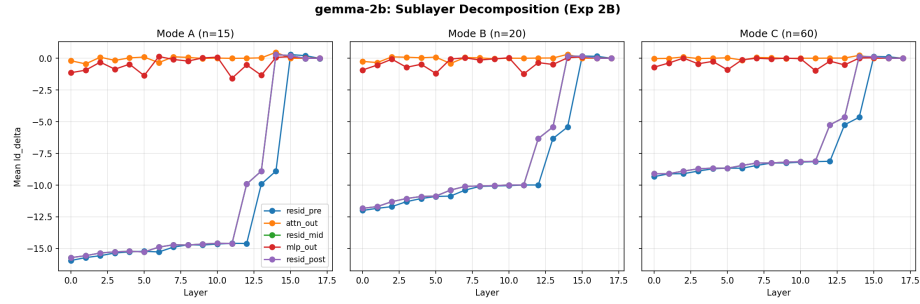


Figure 1: Sublayer decomposition (Exp 2B): `resid_pre` and `resid_post` accumulate damage monotonically across layers 0–14, while `attn_out` and `mlp_out` remain near zero throughout, implicating the residual stream rather than individual sublayers as the primary information carrier.

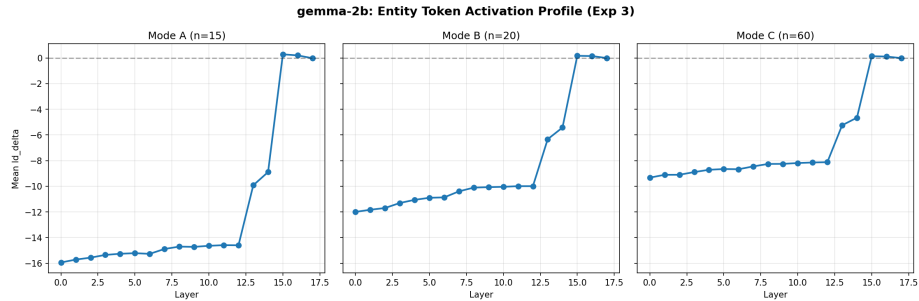


Figure 2: Entity token activation profile (Exp 3), Gemma-2B: patching at the entity token causes substantial logit damage in early and mid layers, followed by a sharp release in the low teens (approximately layers 13–15), with values near zero by the final layers. This supports a handoff from entity-position storage to late-layer/final-token readout rather than a null effect.

3.2 Gemma 12B: Cross-scale replication (mean LD curves, worst layer, confidence vs damage, release histogram, max damage vs release)

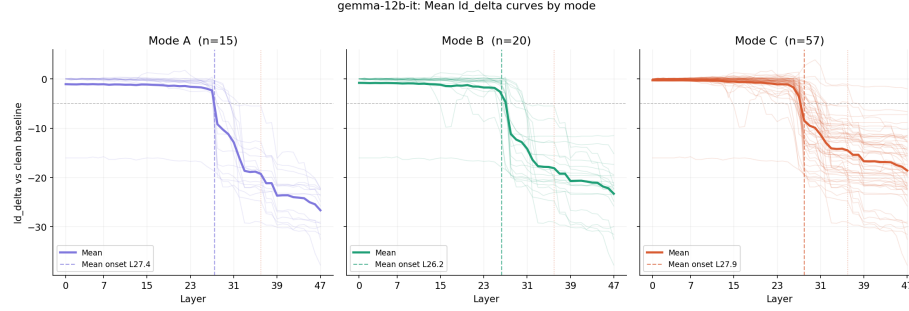


Figure 3: Mean LD_δ curves by mode: damage onset concentrated at layers 26–28 (mean onset: $L_{27.4}$, $L_{26.2}$, $L_{27.9}$ across Modes A/B/C), proportionally consistent with the layer-15 handoff observed in Gemma 2B scaled to 48 layers.

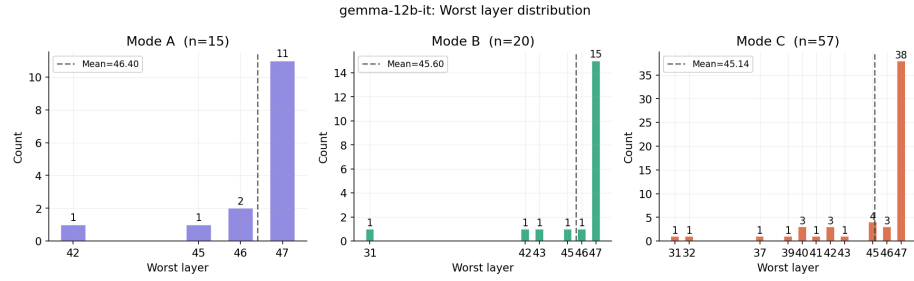


Figure 4: Worst layer distribution: peak damage concentrates at layers 45–47 across all modes (means: 46.40, 45.60, 45.14), mirroring the late-layer readout concentration seen in Gemma 2B.

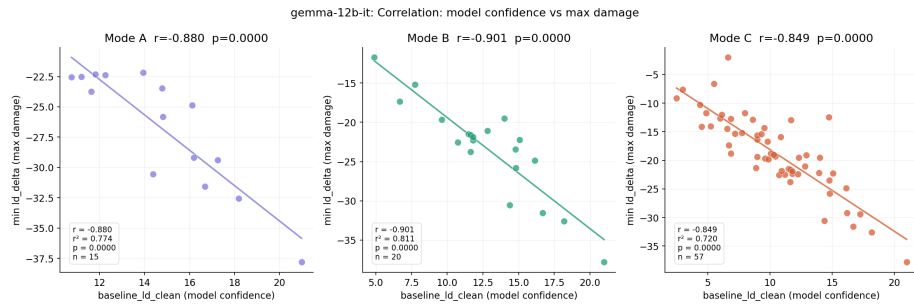


Figure 5: Model confidence vs. max damage: strong negative correlation ($r = -0.880$, -0.901 , -0.849 ; $p < 0.001$ across modes), replicating the Gemma 2B pattern at larger scale.

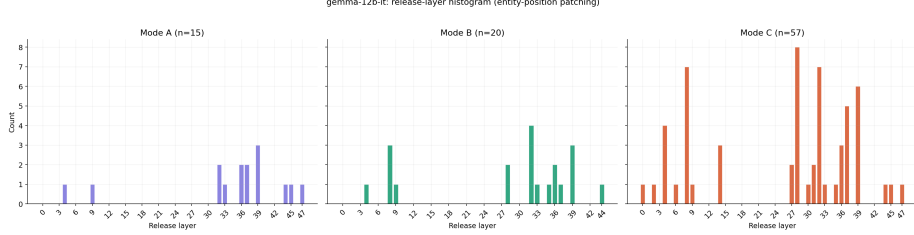


Figure 6: Release layer histogram (Exp 3): signal release is broadly distributed across layers 27–42 in Gemma 12B, contrasting with the sharp layer-15 handoff in Gemma 2B and suggesting a more gradual encoding-to-readout transition at scale.

3.3 Head routing: 2B vs 12B

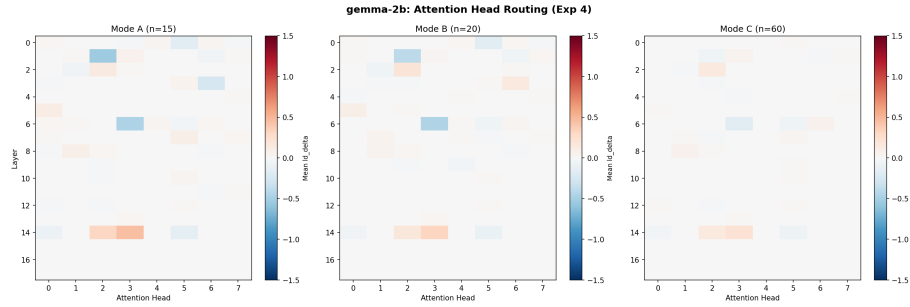


Figure 7: Attention head routing, Gemma 2B (Exp 4): sparse damage pattern concentrated in layers 0–14, with Head 2 and Head 3 at layers 6 and 14 showing the strongest consistent signal across modes.

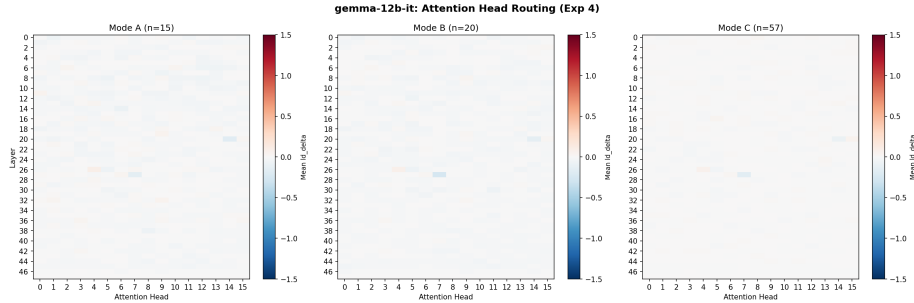


Figure 8: Attention head routing, Gemma-12B-it (Exp 4): compared to Gemma-2B, head-level causal effects are lower and less concentrated (smaller maximum mean cell effect and weaker worst-head concentration), indicating attenuated, more distributed routing at the head level.

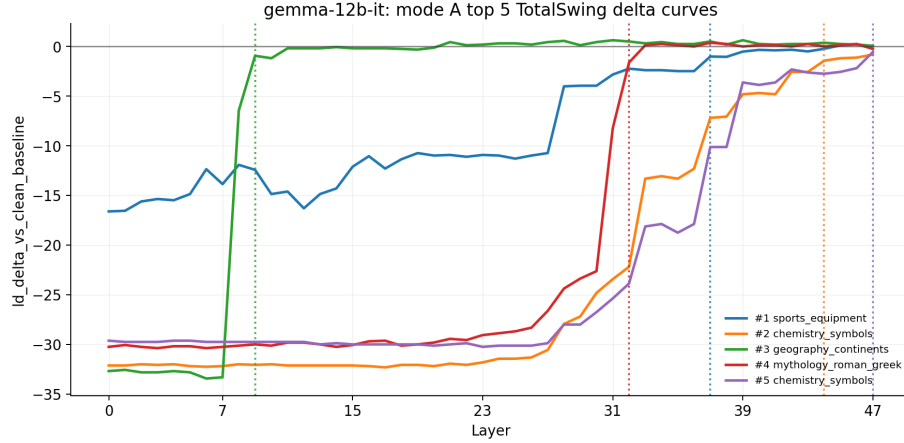


Figure 9: Mode A top-5 TotalSwing pairs by category (Gemma 12B): heterogeneous release profiles reveal that `geography_continents` releases early ($L \approx 8$) while `chemistry_symbols` and `mythology_roman_greek` release sharply near $L=32$, suggesting category-level variation in encoding depth that a single mean onset obscures.

4 Discussion

In this work, I used activation patching to map where factual signal is carried in the model, and the dominant pattern was consistent: the residual stream acts as the main transport pathway, while attention-head interventions produce smaller and more distributed effects. Across experiments, this “plumbing” picture held repeatedly, with the residual stream doing most of the measurable work in preserving or damaging the clean-vs-corrupt logit margin.

A key practical finding is that model-tokenizer behavior is not a minor implementation detail; it changes the experimental substrate itself. Even within the Gemma family (moving from gemma-2b to gemma-12b), I saw 3 golden prompt pairs removed during triage due to tokenizer/compatibility constraints. The same issue was more pronounced when porting to Llama-70B, where 22 prompt pairs were removed. This indicates that cross-model mechanistic comparisons are partly constrained by tokenizer-induced dataset drift, and reported differences should be interpreted with that caveat in mind.

At the mechanistic level, the reduced prominence of individual attention heads in gemma-12b-it is an interesting scaling result: the model appears to rely less on a few dominant heads and more on diffuse computation, while still exhibiting the same residual-stream-centered pipeline. A natural next step is to pair these causal interventions with Sparse Autoencoders (SAEs) to resolve which features are being transmitted in the residual stream and how those feature circuits differ between gemma-2b, gemma-12b-it, and larger families such as Llama. This

would turn the current “where” map into a more precise “what” and “how” account.

5 Future Work

A central limitation of the present work is that I was unable to execute the planned cross-architecture replication suite because the HPC cluster enforces a strict 30 GB disk quota. The full design intended to extend the activation-patching protocol to five additional model families: LLaMA-70B (8-bit quantized), LLaMA-32B, Gemma-31B, and Gemma-27B, with two comparison axes.

- First, I had planned to compare quantized models (NF4 / INT8 via bit-sandbytes) within a single group to control for quantization-induced activation drift and keep precision conditions consistent
- Second, I wanted to compare fine-tuned variants against their base counterparts using CMAP[6], testing whether the three-phase factual-recall circuit is preserved or reorganized by instruction tuning (QLoRA for quantized models, full LoRA for non-quantized models)
- My intention was to extend this node-level activation patching pipeline to full path patching[4], enabling edge-level causal tests between specific sender and receiver components. This would let me resolve whether the distributed head effects we observe reflect true multi-path routing or cancellation between mixed-sign pathways.

Preliminary work on LLaMA-70B was also started, and the only practical challenge encountered was disk space, not model execution. Inference through the quantized 70B model was working well, and I was able to run forward passes successfully once the tokenizer was regenerated for LLaMA-70b using the existing code path. In other words, tokenizer alignment was handled cleanly by the pipeline, leaving the 30GB storage quota as the real blocker for the cross-model replication suite.

I therefore consider the cross-model replication suite to be the highest-priority extension of this work.

6 Conclusion

I presented a causal tracing study of factual computation using controlled clean-corrupt prompt pairs and activation patching across four experiments, and found a consistent mechanistic picture: the residual stream is the primary pathway carrying task-relevant factual signal, while attention-head interventions are generally smaller and more distributed. This pattern held across both `gemma-2b` and `gemma-12b-it`, with scaling preserving the overall “plumbing” structure but reducing concentrated single-head effects in the larger model.

I also showed that tokenizer behavior materially affects what can be measured even within one model family: triage removed 3 golden pairs when moving from `gemma-2b` to `gemma-12b-it`, and 22 pairs in the Llama-70B transfer setting, highlighting a key cross-model comparability constraint. Together, these results provide both a practical methodology for reproducible mechanistic audits and a forward path toward feature-level analyses (e.g with sparse autoencoders) that can explain not only where information flows, but what representations are being routed as models scale.

References

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